

Уравнения с малым параметром при производных

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In[210]:=

```
ClearAll["Global`*"]
```

Условие задачи

In[211]:=

```
Eq = -μ^2 * D[y[x], {x, 4}] + D[y[x], {x, 2}] + q[x] == 0
```

Out[211]=

$$q[x] + y''[x] - \mu^2 y^{(4)}[x] == 0$$

Общее решение однородного уравнения

In[212]:=

```
Sol0 = DSolve[Eq /. q[x] -> 0, y[x], x][[1]] /. μ^2 -> 1;  
y0 = Sol0[[1, 2]]
```

Out[213]=

$$e^{x/\mu} c_1 + e^{-\frac{x}{\mu}} c_2 + c_3 + x c_4$$

Частное решение неоднородного уравнения

Разлагаем решение по степеням параметра $\mu \rightarrow 0$

In[214]:=

```
g = Sum[y[i][x] * μ^i, {i, 0, 4}]
```

Out[214]=

$$y[0][x] + \mu y[1][x] + \mu^2 y[2][x] + \mu^3 y[3][x] + \mu^4 y[4][x]$$

In[215]:=

```
EqG = Collect[Eq /. Table[D[y[x], {x, i}] → D[g, {x, i}], {i, 0, 4}], μ]
```

Out[215]=

$$q[x] + y[0]''[x] + \mu y[1]''[x] + \mu^2 (y[2]''[x] - y[0]^{(4)}[x]) + \\ \mu^3 (y[3]''[x] - y[1]^{(4)}[x]) + \mu^4 (y[4]''[x] - y[2]^{(4)}[x]) - \mu^5 y[3]^{(4)}[x] - \mu^6 y[4]^{(4)}[x] = 0$$

Находим асимптотическую последовательность (исключая линейную составляющую)

In[216]:=

```
EqY0Y1 = Thread[CoefficientList[EqG[[1]], μ] == 0][[1;;5]];
TableForm[Simplify[EqY0Y1]]
```

Out[217]//TableForm=

$$\begin{aligned} q[x] + y[0]''[x] &= 0 \\ y[1]''[x] &= 0 \\ y[2]''[x] &= y[0]^{(4)}[x] \\ y[3]''[x] &= y[1]^{(4)}[x] \\ y[4]''[x] &= y[2]^{(4)}[x] \end{aligned}$$

In[218]:=

```

SolY[0] = DSolve[EqY0Y1[[1]], y[0][x], x][[1]] /. {C[1] → 0, C[2] → 0};
SolY[1] = DSolve[EqY0Y1[[2]], y[1][x], x][[1]] /. {C[1] → 0, C[2] → 0};
SolY[2] = DSolve[EqY0Y1[[3]] /. Flatten[Table[D[SolY[0], {x, i}], {i, 0, 4}]], y[2][x], x][[1]] /. {C[1] → 0, C[2] → 0};
SolY[3] = DSolve[EqY0Y1[[4]] /. Flatten[Table[D[SolY[1], {x, i}], {i, 0, 4}]], y[3][x], x][[1]] /. {C[1] → 0, C[2] → 0};
SolY[4] = DSolve[EqY0Y1[[5]] /. Flatten[Table[D[SolY[2], {x, i}], {i, 0, 4}]], y[4][x], x][[1]] /. {C[1] → 0, C[2] → 0};

```

In[223]:=

```
Sol1 = Flatten[Table[SolY[i], {i, 0, 4}]]
```

Out[223]=

$$\left\{ y[0][x] \rightarrow \int_1^x \int_1^{K[2]} -q[K[1]] \times dK[1] dK[2], y[1][x] \rightarrow 0, y[2][x] \rightarrow -q[x], y[3][x] \rightarrow 0, y[4][x] \rightarrow -q''[x] \right\}$$

Искомое частное решение

In[224]:=

```
y1 = g /. Sol1
```

Out[224]=

$$-\mu^2 q[x] + \int_1^x \int_1^{K[2]} -q[K[1]] \times dK[1] dK[2] - \mu^4 q''[x]$$

Общее решение неоднородного уравнения

In[225]:=

```
Print["y(x) = ", y0 + y1]
```

$$y(x) = e^{x/\mu} c_1 + e^{-\frac{x}{\mu}} c_2 + c_3 + x c_4 - \mu^2 q[x] + \int_1^x \int_1^{K[2]} -q[K[1]] \times dK[1] dK[2] - \mu^4 q''[x]$$